

Report R18 — two conservation laws, or frozen domain walls? Rules 108/201 and 156/198

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

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Abstract

Hilbert-space fragmentation is often explained by the interaction of two conservation laws — charge and dipole moment, or domain walls and magnetisation (Khemani, Hermele and Nandkishore, arXiv:1912.04300). We test that account against the four rules R9 places off both axes of its map. The verdict is that the fragmentation is **fully constructive from frozen domain walls**, and that the conservation laws, which do exist and are exactly of the promised type, are *consequences* of the wall grammar rather than an independent ingredient. In detail: (i) the conserved densities are perfectly diagonal in the computational basis — $b_i = (1 - Z_i Z_{i+1})/2$ — and what fails is *range*, not basis: there is no single-site charge, so magnetisation is not conserved, and separately the *dipole* $P = \sum_i i b_i$ is not conserved either, which is the real reason the charge/dipole mechanism does not apply here; (ii) there is exactly one conserved domain-wall number per rule, and the two families differ in which one — W_{108}/W_{201} conserve the *staggered* $S = \sum_i (-1)^i b_i$ while W_{156}/W_{198} conserve the *total* $D = \sum_i b_i$, with the other broken in each case; (iii) there are exactly two independent local charges, the sublattice-resolved wall counts $|W_{\text{even}}|$ and $|W_{\text{odd}}|$, whose joint level sets number 36 at $N = 14$ against 3329 Krylov sectors — short of the required cardinality by an exponentially growing factor; (iv) each rule has exactly *one* minimal wall word — 00, 11, 01, 10 — and the *set of its occurrences* is a complete invariant, verified exhaustively to $N = 14$; (v) the wall partition reproduces not merely the sector count but the entire multiset of sector sizes, so both of R9's exponents fall out of it, $\rho^2 = 1.754878$ from an exact order-3 integer recurrence and φ from Fibonacci; and (vi) the conserved charge is a *function* of the wall set — for the chiral pair simply $D = 2|W|$ — so it carries strictly less information than the walls do.

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1 The four rules and the question

All four rules are unitary, so by R17 their terminal SCCs are their Krylov sectors, and all four carry a single Hadamard whose gate fires on one neighbour pattern:

rule	tuple	fires when $(x_{i-1}, x_{i+1}) =$	family
W_{201}	VIII	(0, 0)	the PXP constraint
W_{108}	IIIV	(1, 1)	its spin-flip image
W_{156}	IIVI	(1, 0)	chiral
W_{198}	IVII	(0, 1)	its reflection

R9 reports sector-count bases $\rho^2 = 1.754878$ for the first pair and φ for the second, with ρ the plastic number. The question is whether that fragmentation is the shadow of two conserved densities, as in the dipole-conserving models, or whether it is generated constructively by frozen walls.

A note on where the tools came from. The wall grammar and the charge null-space detector are Tier-2 modules (`qca/walls.py`, `qca/charges.py`); they were run read-only for this report. The wall detector in `scaling/wall_charges.py` is an *independent* Tier-1 reimplement, and a regression test pins its output against the four words Tier-2 reports, so the central claim rests on two separate implementations agreeing. The two engines were also checked against each other on the sector counts and largest-sector sizes of all four rules across $N = 6 \dots 14$: they agree term for term.

2 The conservation laws exist, and they are domain-wall numbers

Write $b_i = x_i \oplus x_{i+1}$ for the bond (domain-wall) variables, and

$$D = \sum_i b_i, \quad S = \sum_i (-1)^i b_i.$$

Proposition 1. *For W_{108} and W_{201} , S is conserved on every sector and D is not. For W_{156} and W_{198} , D is conserved and S is not. This holds at both *obc0* and *pbc* for every N tested.*

2.1 These are computational-basis charges; what fails is range

It is worth being careful here, because the loose phrase “bond basis” invites a wrong reading. Everything in this report is diagonal in the computational basis. The bond variable is

$$b_i = \frac{1}{2}(1 - Z_i Z_{i+1}),$$

a two-site diagonal operator, exactly as in the rule-150 domain-wall count of R8; and the entire sector decomposition is a statement about basis states, so nothing off-diagonal could enter. Bond variables are a change of *variables*, not of basis, and Tier-2’s `bond=True/False` flag merely selects which local variables the charge ansatz is built from.

What is actually true is a statement about **range**. Tier-2's detector finds an empty null space at range 1: there is no conserved quantity of the form $\sum_i q(x_i)$, so magnetisation is not conserved — immediate once one notices that the gate flips a site outright. The conserved densities are range 2, and they are certified as such *directly in the site basis*: each of the four rules has a six-dimensional space of certified range-2 site-basis charges, and D (chiral pair) or S (PXP-like pair) is among them, written as a function of the two-site pattern (x_i, x_{i+1}) .

2.2 The dipole is the thing that is missing

The charge/dipole mechanism needs a conserved density *and* a conserved first moment of it. The first exists here. The second does not:

Proposition 2. $P = \sum_i i b_i$, the dipole moment of the domain-wall density, is not conserved for any of the four rules, at either boundary condition.

The reason is transparent from the moves. A single flip of x_i toggles the two bonds b_{i-1} and b_i ; on the sectors where the wall number is conserved they toggle oppositely, so a wall *hops by one site*, leaving D fixed and shifting P by ± 1 . Forbidding exactly that hop is what a dipole-conserving model does, and it is what these rules do not do. So the correct statement is not that a diagonal charge is missing, but that the conserved wall number comes without a conserved dipole.

2.3 Exactly two independent charges, and they are wall counts

The rank of the certified charge values on the full 2^N space is 3 including the constant, i.e. there are exactly **two** independent local conserved quantities per rule — the right number for the two-conservation-law story, which is precisely why the story deserves testing rather than dismissal. They can be named: the **sublattice-resolved wall counts**

$$|W_{\text{even}}| = \#\{i \in W : i \text{ even}\}, \quad |W_{\text{odd}}| = \#\{i \in W : i \text{ odd}\},$$

for all four rules and both boundary conditions, which is why the detector needs its parity split to see two of them. Every other charge in play is a function of this pair: $D = 2(|W_{\text{even}}| + |W_{\text{odd}}|)$ for the chiral rules, and S is their difference up to sign. Their joint level sets reproduce Tier-2's full certified charge set exactly — 13, 18, 25 at $N = 8, 10, 12$ (pbc) for W_{108} and 12, 17, 23 for W_{156} — which is the cross-check that nothing conserved has been left out of the comparison below.

Note already what this says: the two conservation laws are themselves wall data, coarse-grained by sublattice.

3 But they are far too coarse to explain the fragmentation

rule	conserved	N	levels (wall charge)	levels (both)	Krylov sectors	ratio
W_{108}	$ W_{\text{even}} , W_{\text{odd}} $	6	7	10	37	3.7
W_{108}	$ W_{\text{even}} , W_{\text{odd}} $	8	9	15	114	7.6
W_{108}	$ W_{\text{even}} , W_{\text{odd}} $	10	11	21	351	16.7
W_{108}	$ W_{\text{even}} , W_{\text{odd}} $	12	13	28	1081	38.6
W_{108}	$ W_{\text{even}} , W_{\text{odd}} $	14	15	36	3329	92.5
W_{201}	$ W_{\text{even}} , W_{\text{odd}} $	6	6	9	21	2.3
W_{201}	$ W_{\text{even}} , W_{\text{odd}} $	8	8	14	65	4.6
W_{201}	$ W_{\text{even}} , W_{\text{odd}} $	10	10	20	200	10.0
W_{201}	$ W_{\text{even}} , W_{\text{odd}} $	12	12	27	616	22.8
W_{201}	$ W_{\text{even}} , W_{\text{odd}} $	14	14	35	1897	54.2
W_{156}	$ W_{\text{even}} , W_{\text{odd}} $	6	4	10	21	2.1
W_{156}	$ W_{\text{even}} , W_{\text{odd}} $	8	5	15	55	3.7
W_{156}	$ W_{\text{even}} , W_{\text{odd}} $	10	6	21	144	6.9
W_{156}	$ W_{\text{even}} , W_{\text{odd}} $	12	7	28	377	13.5
W_{156}	$ W_{\text{even}} , W_{\text{odd}} $	14	8	36	987	27.4
W_{198}	$ W_{\text{even}} , W_{\text{odd}} $	6	4	10	21	2.1
W_{198}	$ W_{\text{even}} , W_{\text{odd}} $	8	5	15	55	3.7
W_{198}	$ W_{\text{even}} , W_{\text{odd}} $	10	6	21	144	6.9
W_{198}	$ W_{\text{even}} , W_{\text{odd}} $	12	7	28	377	13.5
W_{198}	$ W_{\text{even}} , W_{\text{odd}} $	14	8	36	987	27.4

The comparison uses *both* independent charges, since testing the hypothesis against one of them would understate the symmetry by a factor of N . For W_{108} the joint level sets of $(|W_{\text{even}}|, |W_{\text{odd}}|)$ number 10, 15, 21, 28, 36 over $N = 6, 8, 10, 12, 14$ — visibly quadratic — while the sectors number 37, 114, 351, 1081, 3329. The ratio is therefore not merely large but *exponentially growing*: 3.7, 7.6, 16.7, 38.6, 92.5. For W_{156}/W_{198} it runs 2.1, 3.7, 6.9, 13.5, 27.4 over the same window.

This is the standard diagnostic of fragmentation — exponentially many Krylov sectors inside polynomially many symmetry sectors — and it is exactly what rules out the two-conservation-law account as an *explanation*. Two charges of this kind can label $O(N^2)$ things. They cannot label ρ^{2N} things. Whatever is doing the shattering is not them.

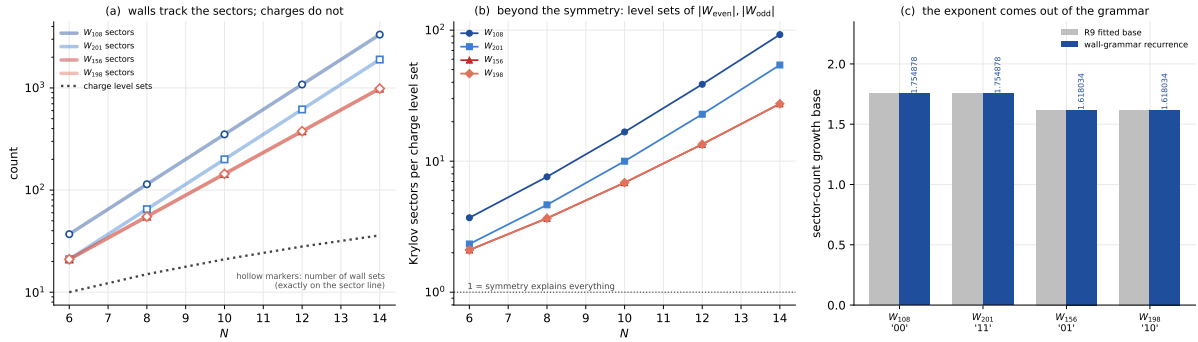


Figure 1: **Walls explain the sectors; the charges cannot.** (a) Sector count (thick line) against the number of distinct wall-occurrence sets (hollow markers) — they coincide exactly at every N — and against the number of joint level sets of the conserved charges (dotted), which stays in single figures. (b) Krylov sectors per symmetry sector: exponentially growing, so the symmetry cannot be the explanation; the dotted line at 1 is where a purely symmetry-resolved model would sit. (c) The dominant root of the exact integer recurrence satisfied by the number of wall sets, against the growth base R9 fitted for the sector count — agreement to six digits, with the exponent derived rather than fitted.

4 The wall grammar is enough

4.1 One word per rule

Detection is the Tier-2 criterion: a word w is a wall if, placed on the chain at either sublattice offset, its sites are invariant under one full cycle for *every* exterior configuration. Since one cycle has light cone 2, two context sites on each side exhaust the exterior dependence and the sixteen configurations of those four sites are the whole test. Each of the four rules has exactly one minimal wall word:

$$W_{108} : 00, \quad W_{201} : 11, \quad W_{156} : 01, \quad W_{198} : 10.$$

For W_{201} this is transparent: a site flips only when both its neighbours are 0, so two adjacent 1s can never move, and neither can anything they flank.

4.2 The occurrence set is a complete invariant

Proposition 3. *Two states lie in the same Krylov sector if and only if the rule’s wall word occurs at the same set of positions. Verified exhaustively for all four rules at **obc0** for $N \leq 14$, and at **pbc** for W_{108} and W_{201} .*

Both halves matter and are reported separately in **wall_completeness**: the wall set is *constant* on each sector (it is conserved) and *injective* across sectors (it separates them). What makes this a constructive account rather than a restatement is that the label is computed from a single state, with no reference to the sector it lies in. By contrast the “frozen-site signature” — the positions and values of the sites that happen to be constant across a sector — is also a complete invariant for all four rules, but it is nearly vacuous, because it is defined *from* the sector. We record it only to note that it is not the content of this section.

rule	tuple	wall	conserved	N	sectors	wall sets	complete	sizes	rec. order	root	R9 base
W_{108}	IIIV	00	S	14	3329	3329	yes	yes	3	1.754878	1.754878
W_{201}	VIII	11	S	14	1897	1897	yes	yes	3	1.754878	1.754878
W_{156}	IIVI	01	D, n01, n10	14	987	987	yes	yes	2	1.618034	1.618034
W_{198}	IVII	10	D, n01, n10	14	987	987	yes	yes	2	1.618034	1.618034

4.3 Both exponents fall out of it

The number of realisable wall sets satisfies an exact homogeneous linear recurrence with integer coefficients, found by exact rational elimination rather than by fitting:

$$\begin{aligned} W_{108}, W_{201} : \quad a(N) &= 2a(N-1) - a(N-2) + a(N-3), & \text{dominant root } 1.754878 &= \rho^2, \\ W_{156}, W_{198} : \quad a(N) &= a(N-1) + a(N-2), & \text{dominant root } 1.618034 &= \varphi. \end{aligned}$$

These reproduce R9’s fitted sector-count bases to six digits (Fig. 1c). Moreover the wall partition gives the *whole* size distribution, not just the count: the multiset of preimage sizes of the map $x \mapsto W(x)$ equals the multiset of sector sizes at every N tested, so $D_{\max}$ comes out too — Fibonacci for W_{108}/W_{201} , and R9’s room-packing series 6, 9, 12, 16, 20, 27, 36, 48, 64, $\dots$ for W_{156}/W_{198} .

4.4 Two details that are not decoration

The boundary is wall material. At **obc0** the chain is padded with a frozen 0 outside each end, and the wall word must be searched on the padded string. This is not bookkeeping:

W_{108} 's word is 00, the padding supplies 0s, and searching the bare chain finds only 616 of the 1081 sectors at $N = 12$. An earlier pass of this analysis concluded from the bare search that W_{156}/W_{198} needed the charge *in addition* to the walls; that was an artefact of the missing padding and is not the case.

One defect, on the ring. For W_{156}/W_{198} at pbc the wall set is complete except for a single collision at every N : the two uniform states 0^N and 1^N are both frozen singletons and neither contains 01, so they share the empty wall set and $n_{\text{sectors}} = n_{\text{wall sets}} + 1$ exactly (48 vs 47, 124 vs 123, 323 vs 322, 844 vs 843). W_{108}/W_{201} have no such defect, because for them 0^N and 1^N are not both frozen.

5 The charge is a shadow of the grammar

The decisive step is not that the walls suffice but that the charges are *downstream* of them.

Proposition 4. *For all four rules, at both boundary conditions and every N tested, the conserved charge is a function of the wall set. For W_{156} and W_{198} it is explicitly*

$$D = 2|W|,$$

$|W|$ being the number of occurrences of the wall word.

The relation for the chiral pair is transparent once stated: on a ring the 01 and 10 occurrences are equinumerous, and D counts both, so the total domain-wall number is exactly twice the number of walls. For W_{108}/W_{201} the staggered charge S is likewise determined by the wall set but is not affine in $|W|$ — it depends on the parities of the wall positions, which is what “staggered” means.

So the implication runs one way only: the wall configuration determines the charge, the charge does not determine the wall configuration, and the sectors are the level sets of the former. The conservation law is a coarse-grained readout of the wall data, not a second ingredient interacting with it.

6 Answering the question as asked

1. **Is it explained by the interaction of two conservation laws?** No. There are exactly two independent local conserved quantities, which is the right number, and they are genuinely domain-wall charges, which is the right kind. But their joint level sets number $O(N^2)$ — 36 of them at $N = 14$ — against 3329 Krylov sectors, and the ratio grows from 3.7 to 92.5 over $N = 6 \dots 14$. They do not have the cardinality to be the explanation. Nor is the missing ingredient a dipole: $\sum_i i b_i$ is not conserved, because the walls hop.
2. **Is it fully constructive from the frozen domain walls?** Yes. One minimal wall word per rule; the set of its occurrences is a complete invariant, computed locally from a single state; the induced partition reproduces the sector count, the full size distribution and both of R9's exponents, with the exponents coming from exact integer recurrences rather than fits. The single exception is the two uniform states on the ring for the chiral pair.
3. **And the relation between the two accounts?** The charges are functions of the wall set — $D = 2|W|$ for the chiral pair — so the constructive account strictly subsumes the symmetry account. This is the opposite of the dipole-conserving picture, where the charges are given and the shattering is an emergent extra; here the walls are given and the charges are the emergent extra.

4. **An incidental contrast worth keeping.** Which wall-number charge survives is exactly what separates the two families: the PXP-like pair conserves the staggered wall number and the chiral pair the total one, and that difference tracks the difference in their exponents (ρ^2 against φ).

7 Reproduction

```
python -m qca_fragmentation.scaling.wall_charges --bc obc0 --rebuild
python -m pytest qca_fragmentation/tests/test_wall_charges.py -q
```

Everything here comes from `analytics/wall_charges_obc0.json`. The Tier-2 detectors used for the cross-checks in §2 — the null-space charge detector, its gate-local certificate, and the wall grammar — live in the Tier-2 repository and were not modified; only read. If this analysis is to be adopted there, `scaling/wall_charges.py` depends on Tier-1 only for `core.cycle.succ`, `core.rules` and the union-find in `graph.wcc.make_succ`, all of which Tier-2 has equivalents for.